Statistical Linear Models Assigment

Patrick Dhatemwa(191894), Ian Mulindwa(192869), Cynthia Musangi(190362), Francis Gitau(189104)

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Objective : Fit a linear model with mpg as the dependent variable and wt and hp as the independent variables.

Loading the mtcars dataset using the code below:

```
data(mtcars)
attach(mtcars)
```

For a better understanding of the data, we view the first 5 rows.

```
head(mtcars)

                    mpg cyl disp  hp drat    wt  qsec vs am gear carb
 Mazda RX4         21.0   6  160 110 3.90 2.620 16.46  0  1    4    4
 Mazda RX4 Wag     21.0   6  160 110 3.90 2.875 17.02  0  1    4    4
 Datsun 710        22.8   4  108  93 3.85 2.320 18.61  1  1    4    1
 Hornet 4 Drive    21.4   6  258 110 3.08 3.215 19.44  1  0    3    1
 Hornet Sportabout 18.7   8  360 175 3.15 3.440 17.02  0  0    3    2
 Valiant           18.1   6  225 105 2.76 3.460 20.22  1  0    3    1
```

To fit the model, we start by defining varibales

```
----------------------------------------
          QUESTION 1 (a)
----------------------------------------


# Set mpg as the dependent variable. (Y)
Y <- (mtcars$mpg)
Y

  [1] 21.0 21.0 22.8 21.4 18.7 18.1 14.3 24.4 22.8 19.2 17.8 16.4 17.3 15.2 1
0.4
 [16] 10.4 14.7 32.4 30.4 33.9 21.5 15.5 15.2 13.3 19.2 27.3 26.0 30.4 15.8 1
9.7
 [31] 15.0 21.4
```

We now go ahead to define the independent variables. (X1 [wt], and X2 [hp])

After defining the independent variables, we then go ahead to find the transpose of the matrix of independent variables to meet our objective.

```
# Set wt and hp as the independent variables. (X1, X2)
data(mtcars)
attach(mtcars)
X <- cbind(wt, hp)
X_t <- t(X) # find the transpose of X.
X_t # Call the function.
```

```
      [,1]    [,2]  [,3]    [,4]   [,5]   [,6]   [,7]  [,8]  [,9]  [,10]  [,1
1]
 wt   2.62   2.875  2.32   3.215   3.44   3.46   3.57  3.19  3.15   3.44   3.
44

 hp 110.00 110.000 93.00 110.000 175.00 105.00 245.00 62.00 95.00 123.00 123.
00
     [,12]  [,13]  [,14]  [,15]   [,16]   [,17] [,18]  [,19]  [,20]  [,21]
 wt   4.07   3.73   3.78   5.25   5.424   5.345   2.2  1.615  1.835  2.465
 hp 180.00 180.00 180.00 205.00 215.000 230.000  66.0 52.000 65.000 97.000
     [,22]   [,23]  [,24]   [,25]  [,26] [,27]   [,28]  [,29]  [,30]  [,31]
 wt   3.52   3.435   3.84   3.845  1.935  2.14   1.513   3.17   2.77   3.57
 hp 150.00 150.000 245.00 175.000 66.000 91.00 113.000 264.00 175.00 335.00
     [,32]
 wt   2.78
 hp 109.00
```

Step 1: Fitting the linear model. (OLS formulation)

```
N <- X_t%*%X
N # Call the function
```

```
            wt        hp
 wt   360.9011  16471.74
 hp 16471.7440 834278.00
```

Step 2: Obtain the inverse of the matrix using the solve function.

```
# Obtain the inverse of the matrix resulting from the product of the X matrix
and its transpose.
solve(N)
```

```
               wt            hp
 wt  0.0280208498 -5.532356e-04
 hp -0.0005532356  1.212156e-05
```

Step 3: Now we obtain the matrix obtained from the product of X-transpose and the Y-matrix

```
f <- X_t%*%Y
f

         [,1]
 wt  1909.753
 hp 84362.700
```

Note: according to OLS, $$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$$

But according to our code, that will be Solve(N) * f

Step 4 : Final OLS formulation.

```
B = solve(N)%*%f
B # Call B

           [,1]
 wt  6.84044997
 hp -0.03393526
```

Step 5 : The linear model.

```
model <- lm(mpg ~ wt + hp, data = mtcars)
show(model)


 Call:
 lm(formula = mpg ~ wt + hp, data = mtcars)


 Coefficients:
 (Intercept)           wt           hp
    37.22727     -3.87783     -0.03177
```

$$\widehat{mpg} = 37.22727 - 3.87783(wt) - 0.03177(hp)$$

Shown above is the fitted linear model.

```r
summary(model)
```

```
Call:
lm(formula = mpg ~ wt + hp, data = mtcars)

Residuals:
   Min     1Q Median     3Q    Max 
-3.941 -1.600 -0.182  1.050  5.854 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept) 37.22727    1.59879  23.285  < 2e-16 ***
wt          -3.87783    0.63273  -6.129 1.12e-06 ***
hp          -0.03177    0.00903  -3.519  0.00145 ** 
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.593 on 29 degrees of freedom
Multiple R-squared:  0.8268, Adjusted R-squared:  0.8148 
F-statistic: 69.21 on 2 and 29 DF,  p-value: 9.109e-12
```

---

i) **P- value** : (9.109e-12) This P-value is less than 0.05, and therefore we can say that our model is significant.

ii) Interpretation of coefficients.

Bo - The minimum value of mpg when hp and (wt) are zero. (i.e. 37.23) which is the minimum level of fuel efficiency.

B1 – holding horsepower constant, a unit increase in (wt) decreases fuel efficiency (mpg) by 3.87783 units. Meaning if the weight of the car increases by about 1000 pounds, full efficiency drops by about 3.88 mpg

B2 - holding weight constant, a unit increase in horsepower decreases (mpg) by 0.03177 units

iii)

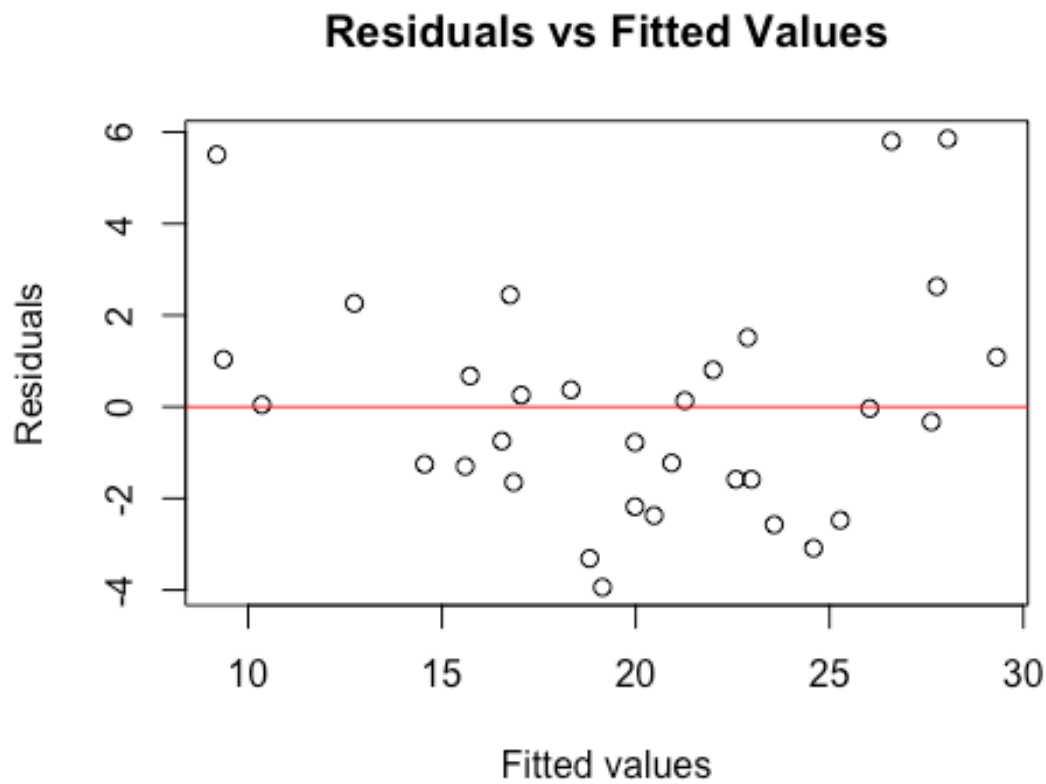
a) **R-squared Value (0.8268)** : This means that 82.68% of the variation in the model can be explained by (wt) and (hp).

b) **Adjusted R-squared value (0.8148)** : This means that 81.48% of the variation in the model can still be explained by (wt) and (hp) meaning that the model is still very significant since the difference between the R-squared and the adjusted R-squared is not very big.

B) REGRESSION ASSUMPTIONS.

```
# i)     Check linearity and interpret the findings.

plot(model$fitted.values, resid(model),
     xlab = "Fitted values",
     ylab = "Residuals",
     main = "Residuals vs Fitted Values")
abline(h = 0, col = "red")
```



Assumption 1: LINEARITY

We are checking to see if residuals are randomly scattered around zero and not having an obvious pattern or curve which violates linearity.

According to the above plot, linearity is not completely violated.

Assumption 2 : Check independence of errors.

```
#install.packages("lmtest")
#install.packages("car")

library(lmtest)
```

```r
library(car)

dwtest(model)


  Durbin-Watson test

 data:  model
 DW = 1.3624, p-value = 0.02061
 alternative hypothesis: true autocorrelation is greater than 0
```

Here we are checking whether residuals are independent of one another.

Hypotheses

H0 : Errors are independent

H1 : Errors are auto-correlated.

Interpretation

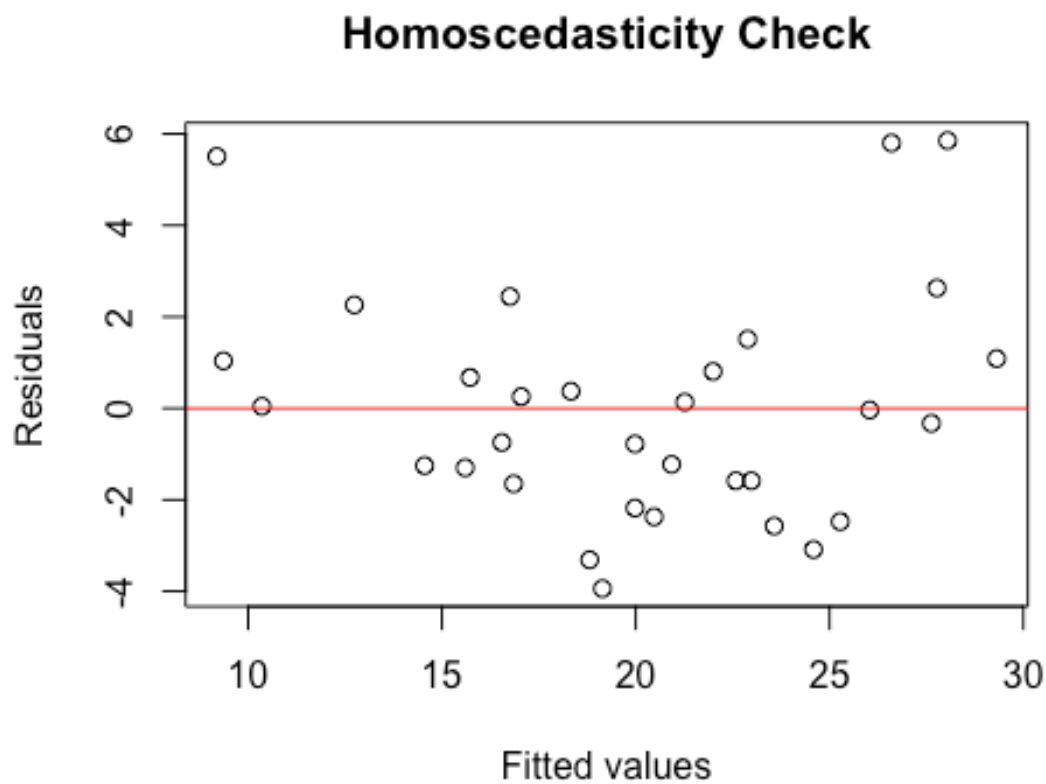
If p-value > 0.05, we fail to reject H0. Independence is acceptable.

If p-value < 0.05, there may be autocorrelation.

But since p-value < 0.05 (i.e. 0.02061), we conclude that there may be auto-correlation.

Assumption 3 : Check constant variance (homoscedasticity).

```r
plot(model$fitted.values, resid(model),
     xlab = "Fitted values",
     ylab = "Residuals",
     main = "Homoscedasticity Check")
abline(h = 0, col = "red")
```



Here we want the residuals to have roughly equal spread across fitted values.

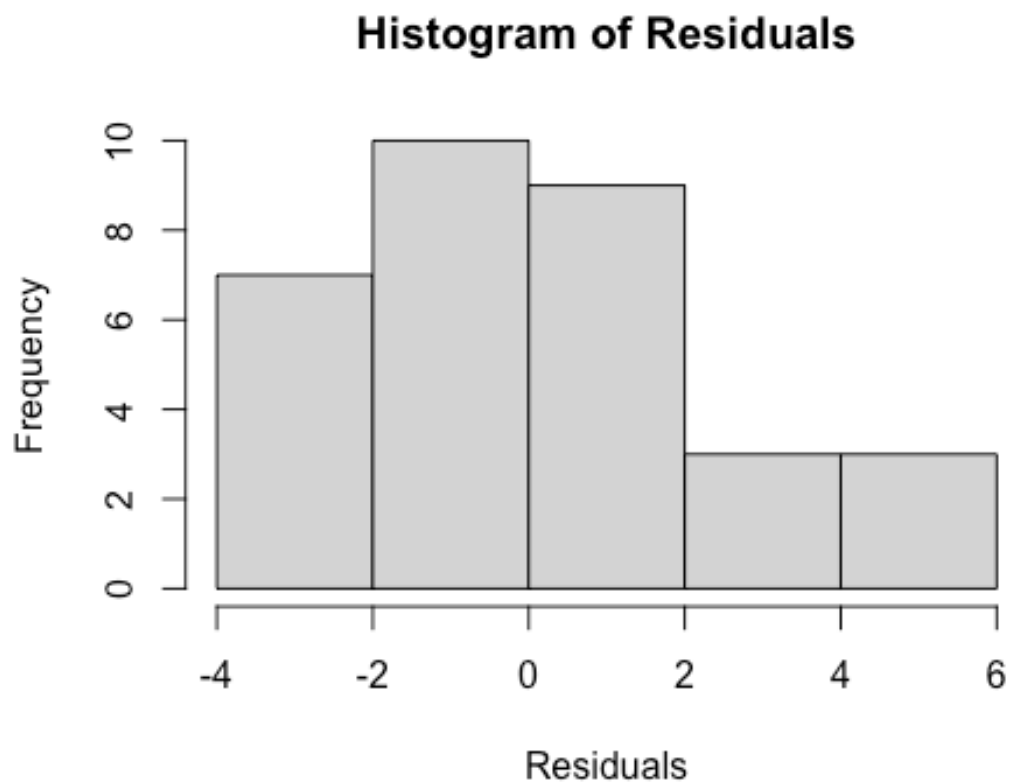
Interpretation

If residuals spread evenly, homoscedasticity is satisfied. If the residuals form a funnel shape, variance is not constant.

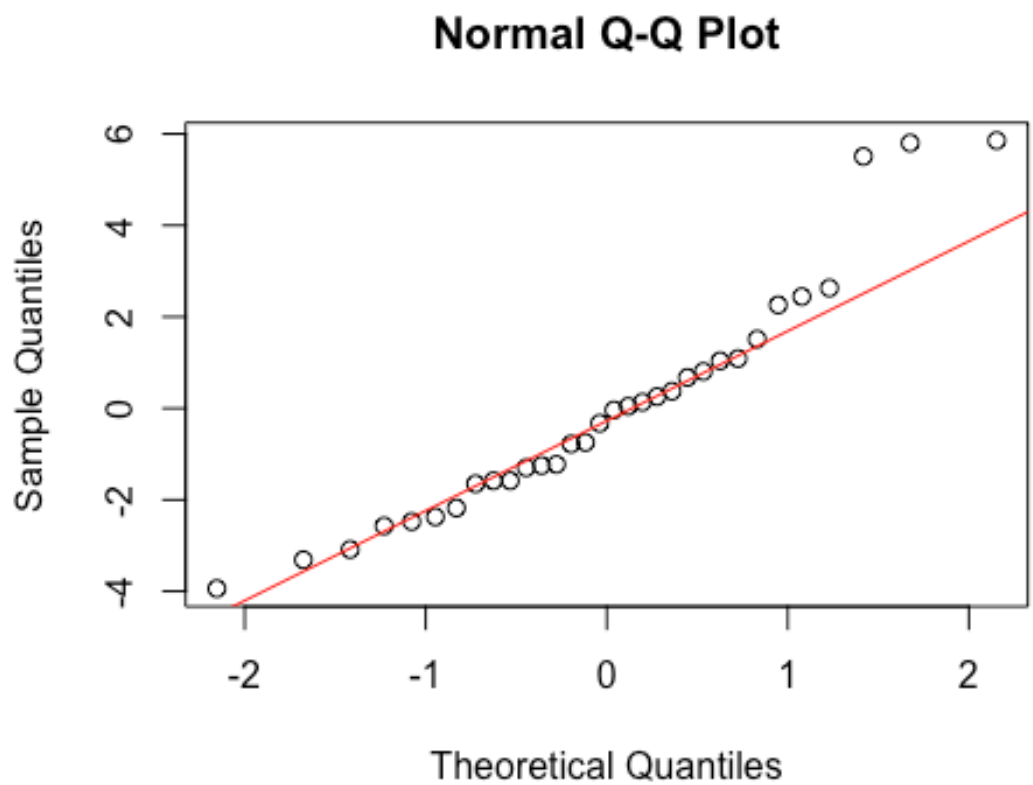
And according to the above plot, homoscedasticity is satisfied, since the residuals spread evenly.

Assumption 4 : Check the normality of residuals.

```
hist(resid(model),
     main = "Histogram of Residuals",
     xlab = "Residuals")
```




```
qqnorm(resid(model))
qqline(resid(model), col = "red")
```




```
shapiro.test(resid(model))


  Shapiro-Wilk normality test

data:  resid(model)
W = 0.92792, p-value = 0.03427
```

The normality of residuals was assessed using a histogram, Q-Q plot, and the Shapiro Wilk test. The Shapiro Wilk test produced a p-value of 0.03427, which is less than 0.05.

Therefore, the null hypothesis of normality was rejected, indicating that the residuals are not perfectly normally distributed. However, the Q-Q plot showed only moderate deviation from the reference line, suggesting that the violation of normality is not severe.

Assumption 5 : Check for multicollinearity.

```
library(car)
vif(model)

       wt       hp
 1.766625 1.766625
```

Here we are checking whether weight, (wt) and horse power, (hp) are too highly correlated.

Interpretation

Common rule: VIF < 5 acceptable VIF 5 to 10 concerning VIF > 10 serious multicollinearities

If VIF values are below 5, multicollinearity is not a serious problem.

Since both values for wt, and hp are < 5, the test outcome is accessible.

C) Perform Case Diagnostics.

(i)Standard residuals

```
std_res <- rstandard(model)
std_res

           Mazda RX4       Mazda RX4 Wag          Datsun 710      Hornet 4 Dr
ive
         -1.01458647         -0.62332752         -0.98475880          0.05332
850
   Hornet Sportabout             Valiant          Duster 360           Merc 2
40D
          0.14644776         -0.94769800         -0.53305899          0.62035
108
            Merc 230            Merc 280           Merc 280C          Merc 45
0SE
          0.32068555         -0.30786160         -0.86081660          0.26776
519
          Merc 450SL         Merc 450SLC  Cadillac Fleetwood Lincoln Continen
tal
          0.10087865         -0.65019507          0.01914207          0.44970
228
   Chrysler Imperial            Fiat 128         Honda Civic      Toyota Coro
lla
          2.35451716          2.33192251          0.44781172          2.37861
784
       Toyota Corona    Dodge Challenger         AMC Javelin          Camaro
Z28
         -1.22297829         -1.30026858         -1.54564189         -0.50973
308
    Pontiac Firebird           Fiat X1-9       Porsche 914-2        Lotus Eur
opa
          0.96380569         -0.13220037         -0.01494999          1.10241
512
      Ford Pantera L        Ferrari Dino       Maserati Bora          Volvo 1
42E
         -0.32270884         -0.48919397          1.11989090         -0.62369
214
```

```r
# To identify possible outliers
which(abs(std_res) > 2)
```

```
 Chrysler Imperial          Fiat 128    Toyota Corolla
                17                18                20
```

Common rule: |standardized residual| > 2 possible outlier |standardized residual| > 3 serious outlier These identify observations whose actual mpg is unusually far from predicted mpg.

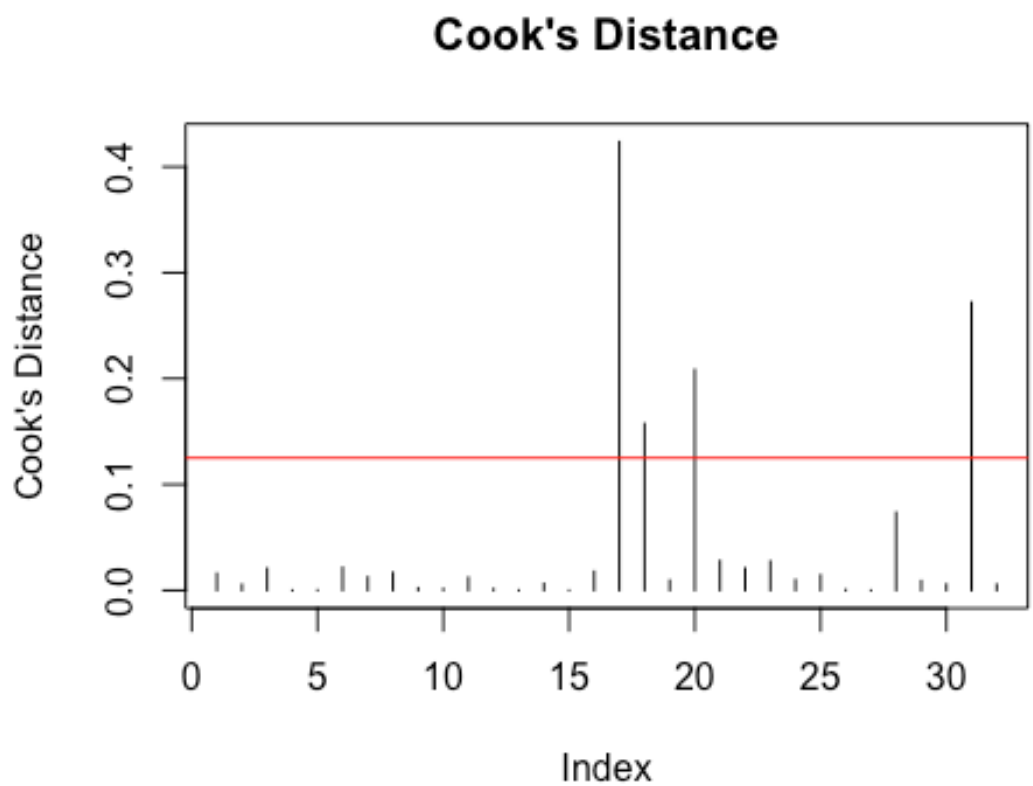
Car Standardized ——- Residual Chrysler Imperial ——- 2.3545 Fiat 128 ——- 2.3319 Toyota Corolla ——- 2.3786

(ii) Cook's Distance

```r
cook <- cooks.distance(model)
cook
```

```
           Mazda RX4       Mazda RX4 Wag          Datsun 710      Hornet 4 Dr
ive
        1.589652e-02        5.464779e-03        2.070651e-02        4.724822e
-05
   Hornet Sportabout             Valiant          Duster 360           Merc 2
40D
        2.736184e-04        2.155064e-02        1.255218e-02        1.677650e
-02
            Merc 230            Merc 280           Merc 280C          Merc 45
0SE
        2.188702e-03        1.554996e-03        1.215737e-02        1.423008e
-03
          Merc 450SL         Merc 450SLC  Cadillac Fleetwood Lincoln Continen
tal
        1.458960e-04        6.266049e-03        2.786686e-05        1.780910e
-02
   Chrysler Imperial            Fiat 128         Honda Civic      Toyota Coro
lla
        4.236109e-01        1.574263e-01        9.371446e-03        2.083933e
-01
       Toyota Corona    Dodge Challenger         AMC Javelin          Camaro
Z28
        2.791982e-02        2.087419e-02        2.751510e-02        9.943527e
-03
    Pontiac Firebird           Fiat X1-9       Porsche 914-2        Lotus Eur
opa
        1.443199e-02        5.920440e-04        5.674986e-06        7.353985e
-02
      Ford Pantera L        Ferrari Dino       Maserati Bora          Volvo 1
42E
        8.919701e-03        5.732672e-03        2.720397e-01        5.600804e
-03
```

```r
plot(cook, type = "h",
     main = "Cook's Distance",
     ylab = "Cook's Distance")
abline(h = 4 / nrow(mtcars), col = "red")
```




```r
# To identify influential cases
which(cook > 4 / nrow(mtcars))
```

```
 Chrysler Imperial          Fiat 128    Toyota Corolla     Maserati Bora
                17                 18                20                31
```

Interpretation.

Cook's distance checks whether one observation strongly affects the regression model.
Rule: Cook's Distance > 4/n

Since n=32:

4/32 = 0.125

Therefore, any case above 0.125 may be influential.

According to our output, The following are influetial to out liear regression model, meaning removing would significantly affect our model.

CAR	COOK'S DISTANCE
Chrysler Imperial	0.4236
Fiat 128	0.1574
Toyota Corolla	0.2087
Maserati Bora	0.2720

Which exceed the threshold of 0.125

Cook's Distance: 0.4236 This is the largest influence in the model. Meaning, the Chrysler Imperial substantially affects the regression estimates. Removing it would noticeably change the fitted regression model.

(iii) Leverage

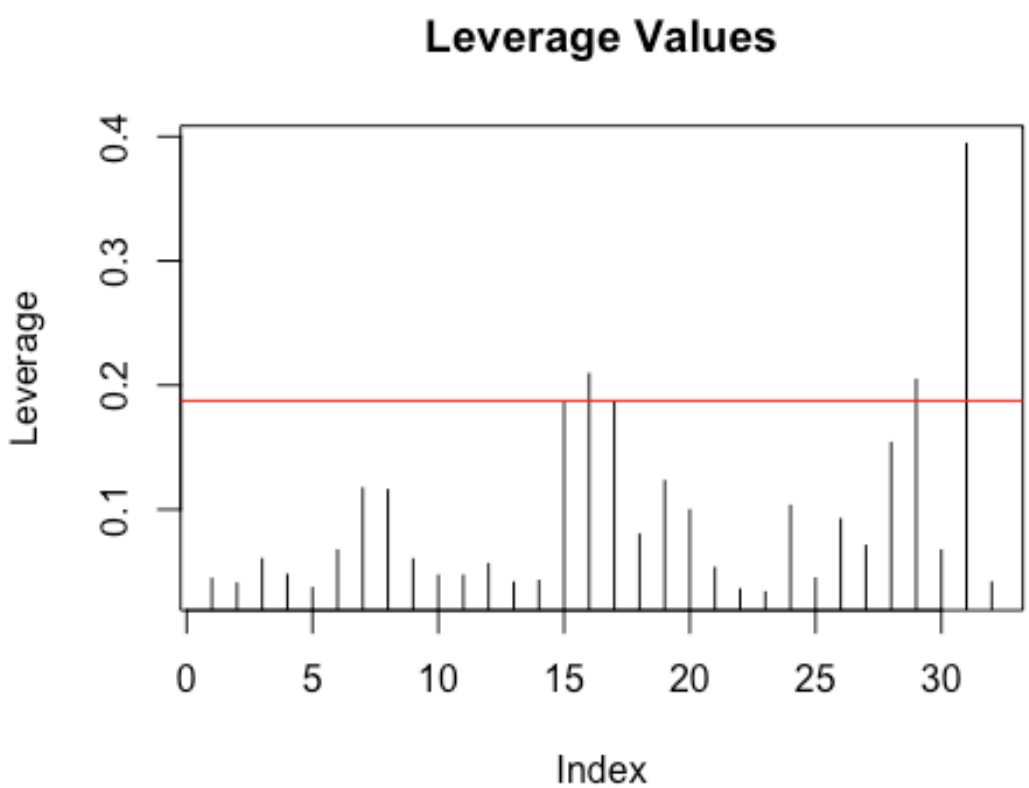
```
lev <- hatvalues(model)
lev

           Mazda RX4       Mazda RX4 Wag          Datsun 710      Hornet 4 Dr
ive
          0.04427691          0.04048669          0.06020097          0.04747
494
   Hornet Sportabout             Valiant          Duster 360           Merc 2
40D
          0.03686288          0.06715104          0.11701542          0.11565
615
            Merc 230            Merc 280           Merc 280C          Merc 45
0SE
          0.06001631          0.04691083          0.04691083          0.05619
559

          Merc 450SL         Merc 450SLC  Cadillac Fleetwood Lincoln Continen
tal
          0.04123610          0.04257292          0.18577112          0.20897
838
   Chrysler Imperial            Fiat 128         Honda Civic      Toyota Coro
lla
          0.18648721          0.07990991          0.12295814          0.09950
```

```
335
       Toyota Corona    Dodge Challenger         AMC Javelin          Camaro
Z28
          0.05303126          0.03571654          0.03339813          0.10298
538
    Pontiac Firebird           Fiat X1-9       Porsche 914-2        Lotus Eur
opa
          0.04453321          0.09225188          0.07078181          0.15364
135
      Ford Pantera L        Ferrari Dino       Maserati Bora          Volvo 1
42E
          0.20442386          0.06704651          0.39420816          0.04140
623
```

```r
plot(lev, type = "h",
     main = "Leverage Values",
     ylab = "Leverage")
abline(h = 2 * 3 / nrow(mtcars), col = "red")
```




```r
# To identify high leverage cases.
which(lev > 2 * 3 / nrow(mtcars))
```

```
 Lincoln Continental      Ford Pantera L       Maserati Bora
                  16                  29                  31
```

Interpretation High leverage observations have unusual predictor values. They may not be outliers in mpg, but their wt and hp combinations are unusual.

(iv) Covarince Ratio

```
cov_ratio <- covratio(model)
cov_ratio

           Mazda RX4       Mazda RX4 Wag          Datsun 710      Hornet 4 Dr
ive
           1.0430373           1.1119752           1.0675097           1.1660
461
   Hornet Sportabout             Valiant          Duster 360           Merc 2
40D
           1.1509803           1.0837277           1.2216245           1.2069
617
            Merc 230            Merc 280           Merc 280C          Merc 45
0SE
           1.1694216           1.1543068           1.0786052           1.1684
570
          Merc 450SL         Merc 450SLC  Cadillac Fleetwood Lincoln Continen
tal
           1.1575798           1.1104048           1.3644478           1.3753
525
   Chrysler Imperial            Fiat 128         Honda Civic      Toyota Coro
lla
           0.7226653           0.6476489           1.2406773           0.6433
805
       Toyota Corona    Dodge Challenger         AMC Javelin          Camaro
Z28
           1.0009058           0.9621726           0.8880991           1.2055
757
    Pontiac Firebird           Fiat X1-9       Porsche 914-2        Lotus Eur
opa
           1.0545996           1.2217128           1.1956172           1.1544
834
      Ford Pantera L        Ferrari Dino       Maserati Bora          Volvo 1
42E
           1.3815003           1.1616192           1.6061878           1.1129
888

# We now check for influental cases
n <- nrow(mtcars)
p <- 3

lower <- 1 - (3 * p / n)
upper <- 1 + (3 * p / n)

lower

 [1] 0.71875
```

```
upper

 [1] 1.28125

which(cov_ratio < lower | cov_ratio > upper)

  Cadillac Fleetwood Lincoln Continental            Fiat 128      Toyota Coro
lla
                  15                  16                  18
20
      Ford Pantera L       Maserati Bora
                  29                  31
```

The covariance ratio checks whether removing a case substantially changes the precision of the model estimates.

So the acceptable range is approximately: 0.71875 to 1.28125

Cases outside this range may be influential. namely:

1) Cadillac Fleetwood

2) Lincoln Continental

3) Fiat 128

4) Toyota Corolla

5) Ford Pantera L

6) Maserati Bora